The need for Watson

After Deep Blue beat Garry Kasparov at the game of chess, it was assumed that the world of computing had reached a point where it could outthink the human brain. But as we all know, the human brain is a much more complex mechanism. Deep Blue was purely analytical and didn't take into account various different aspects of intelligence. Indeed, it couldn't even dream of passing the Turing Test.

Yet, natural language processing systems were becoming the need of the hour as companies increasingly captured critical business information in natural language documentation. There's growing interest in workload optimised systems that deeply analyse the content of natural language questions to answer those questions with precision. Thus was born the need for a question-and-answer robot in the form of Watson.

As its maker IBM puts it, "Advances in question answering (QA) technology will increasingly help support professionals in critical and timely decision making in areas such as health care, business intelligence, knowledge discovery, enterprise knowledge management, and customer support."

So how would Watson be judged? IBM wanted a system that would take a test as gruelling and human-centric as Deep Blue's match against Kasparov. The conclusion was to make a system that would effectively beat humans on the game show Jeopardy, where contestants are given an answer and they have to come up with the best question to ask that satisfies the answer.

The game pits three human contestants against one another to answer rich natural language questions over a broad range of topics, with penalties for wrong answers. In this three-person competition, confidence, precision and answering speed are of critical importance, as contestants usually come up with their answers in the few seconds it takes for the host to read a clue. To compete in this game at human-champion levels, a computer system would need to answer roughly 70 per cent replace all of the questions asked with greater than 80 per cent precision in three seconds or less.

"The essence of making decisions is recognising patterns in vast amounts of data, sorting through choices and options, and responding quickly and accurately," said Samuel J. Palmisano, IBM Chairman, President and Chief Executive Officer. "Watson is a compelling example of how the planet – companies, industries, cities – is becoming smarter. With advanced computing power and deep analytics, we can infuse business and societal systems with intelligence."